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RESEARCH

A long-term and pragmatic numerical simulation of wire arc additively manufactured ER70S6 and ER316L thin-wall via finite difference method

Habib Hamed Zargari[1] · Seyyed Ahmadreza Khajeh Mohammadilar[1] · Vahid Ziaei Laleh[1]

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Abstract

A three-dimensional model utilizing the finite difference method (FDM) was developed to visualize the deposition phenomena in the wire arc additive manufacturing (WAAM) process. The presented computational fluid dynamics (CFD) model incorporates heat and mass transfer, droplet impingement, gravity, solid–liquid phase changes, and fluid flow equations, facilitating the prediction of thin-wall formation prior to experimental manufacturing, which is an inherent limitation of conventional Finite Element-based Methods (FEM). Furthermore, in contrast to prior FDM-based models, which have typically been constrained to the limited timeframes, the proposed approach enables realistic process simulation by using simplification assumptions under conventionally applied low-current and extended deposition time in a novel implementation. The established model employs the surface tension temperature-dependent equation as the main controlling factor in boundary conditions, and it can fairly identify the most crucial deposited layer that is sensitive to sagging by integrating the energy, mass, momentum, and modified volume of fluid (VOF) equations. The numerical model has been validated through 25 successive deposits of ER70S6 and ER316L wire steel layers over 3400 s, maintaining an average margin of error within ten percent. Based on the simulation results, the critical surface tension pressure could serve as a key parameter for assessing manufacturing stability and detecting the potential risk of sagging due to its simultaneous dependence on both temperature and the viscosity of molten material. This adaptability has the potential to significantly reduce defects and foster a more sustainable manufacturing environment.

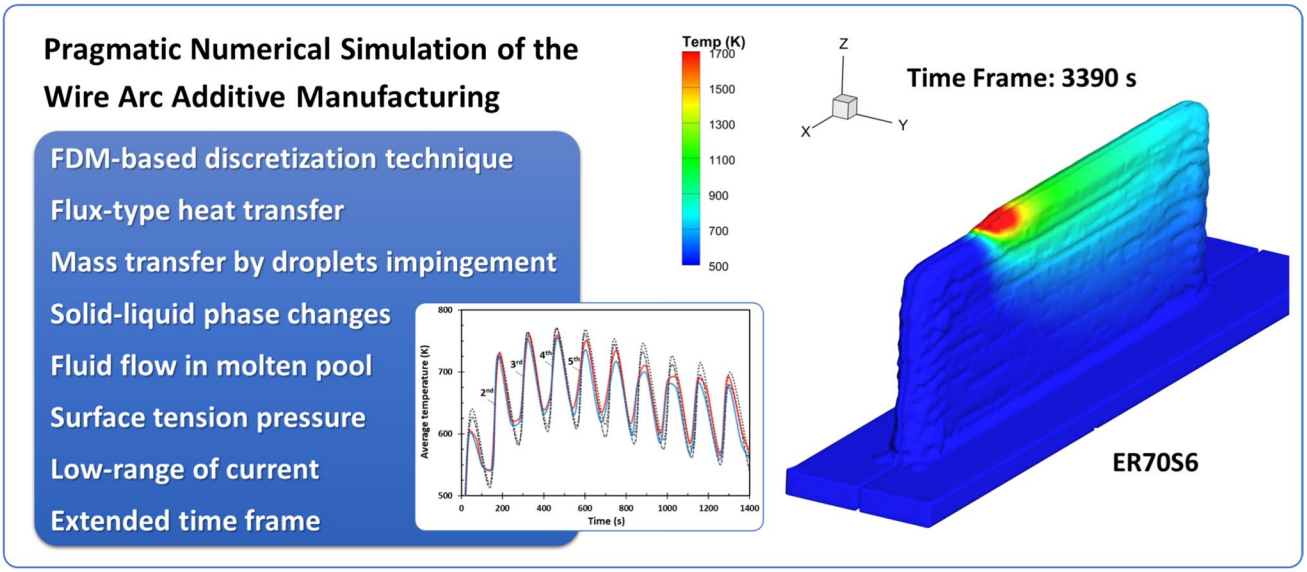
✉ Habib Hamed Zargari
hhamedzargari@sut.ac.ir

Seyyed Ahmadreza Khajeh Mohammadilar
sa_mohammadilar402@sut.ac.ir

Vahid Ziaei Laleh
v_ziaeilaleh403@sut.ac.ir

[1] Advanced Materials Research Institute, Faculty of Materials Engineering, Sahand University of Technology, Tabriz, Islamic Republic of Iran

Graphical abstract



Keywords Finite difference method · Wire arc additive manufacturing · WA-DED · Surface tension · Computational fluid dynamic · ER316L

1 Introduction

Wire arc additive manufacturing (WAAM), which has recently been reported as wire-arc directed energy deposition (WA-DED), holds a prominent position in the additive manufacturing (AM) of medium to large-scale industrial parts due to its commonalities with conventional and economical arc welding procedures. WAAM utilizes arc welding techniques, such as gas metal arc welding (GMAW) and cold metal transfer (CMT), to deposit material layer by layer from a wire feedstock. With high deposition rates and flexibility in material choice, WAAM has become increasingly popular in the aerospace, automotive, and shipbuilding industries, where the demand for large-scale metal components is substantial [1–5].

Among the various materials utilized in the WAAM process, ER70 steel and ER316L stainless steel wires stand out due to their satisfactory mechanical properties and suitability for various applications. The distinctive properties of ER70 steel wire, including its exceptional weldability, and those of ER316L stainless steel, such as its superior corrosion resistance, combined with the advantages of WAAM technology, offer a significant opportunity to improve manufacturing efficiency and minimize material waste. However, the optimal process parameters for WAAM are critical for achieving the desired mechanical properties. Studies on ER316L show that wire feed speed (WFS) and torch travel speed (TS) significantly influence the mechanical characteristics of the manufactured components [6]. As is clear, one of the challenges in WAAM is controlling heat input, which affects the microstructure, mechanical properties, and quality of the manufactured part [7]. Excessive heat can cause defects like keyhole porosity and grain growth, while insufficient heat can lead to poor fusion between layers [8]. Optimizing heat input is crucial, and techniques such as Fronius Cold Metal Transfer (CMT), Miller Regulated Metal Deposition (RMD™), and Merkle ColdMIG have been explored to address this issue utilizing the specific pulsed mode of current. Pulsed gas metal arc welding (P-GMAW), widely used in WAAM, enables controlled heat input by specific pulsing the current, resulting in smaller droplets, smoother surfaces, and reduced residual stresses. However, P-GMAW requires precise control of multiple parameters, making robust process models and optimization techniques essential for its effective use [9, 10]. Ríos et al. [11] examined conditions that generated identical deposit geometries despite significant variations in the process parameters. It was observed that the parameters leading to a depression in the molten pool demanded reduced power, potentially attributed to enhanced melting efficiency and convection effect.

Looking forward, the ongoing development of WAAM technology will likely focus on several key areas. First, the implementation of real-time monitoring and control systems will be crucial to improving process stability and ensuring consistent part quality. Second, the incorporation of sophisticated simulation models, using machine learning and

artificial intelligence, holds promise for optimizing WAAM processes through real-time data analysis, reducing the reliance on trial-and-error methods [12]. Finally, expanding WAAM to new materials and applications. Numerical modeling of the WAAM processes supports the advancement of the quality of manufacturing parts and provides valuable guidance to predict the behavior of the deposited layers. Over the past three decades, numerical simulations for the single-pass GMAW have seen significant advancements. Cho et al. [13] introduced a model to represent the dynamics of the weld pool in P-GMAW, later expanding this model to predict the formation of weld bead humps [14]. Cheon et al. [15] explored the impact of different simulation methods on the thermal behavior of GMAW, which they simulated and visualized. Wu et al. created numerical models for hybrid GMAW processes [16] using comprehensive computational fluid dynamics (CFD) based simulations. Weld pool dynamics in P-GMAW are visualized in detail considering the surface tension active element [17]. Ongoing advancements in numerical modeling continue, particularly in areas such as WAAM [18, 19], where new industrial applications are continually emerging. The Volume of Fluid (VOF) technique, introduced by Hirt and Nichols [20], has been widely used to track the free surface of the puddle in welding simulations, with several improvements made over time to enhance accuracy [21–23]. The subsequent improvements, such as Lagrangian-based VOF transfer, have significantly enhanced the precision of these models, though further refinements are needed for specialized applications like hybrid welding [24].

Despite these advancements, the complex thermal cycles in WAAM continue to pose challenges in achieving uniform microstructures and mechanical properties across the entire component. The repeated heating and cooling cycles during the layer-by-layer deposition process can lead to significant thermal gradients, which in turn cause residual stresses and anisotropic properties in the final product. To address these issues, researchers have developed various simulation models to predict and control the thermal behavior of the WAAM process. Finite Element Method (FEM) has been widely used to simulate heat transfer and solidification during the deposition process, providing insights into the influence of process parameters on the quality of the final part [25, 26]. These models allow for the optimizing heat input, cooling rates, and inter-pass temperatures, which are needed for minimizing residual stresses and controlling the microstructure. However, conventional FEM-based analyses have been unable to accurately predict the thickness, thin-wall shape, and formation possibility prior to experimental sample production, relying solely on the law of mass conservation and thermal analysis [27]. This method essentially utilizes element birth and death meshing techniques, which do not align entirely with the physical conditions of droplets generated in the welding process and the subsequent effect of droplets' impact on the weld geometry. Briefly, the entire thin-wall is produced uniformly without considering the geometry changes resulting from previous layers. In this regard, studies have been conducted by La Peña et al. [28], in which they assumed that a semi-circular uniform bead approximation provides better results than a rectangular one. Furthermore, Aryal et al. [29] investigated how ignoring the free surface deformation affects the maximum temperature and fluid velocity in simulating the gas metal arc welding process.

One of the critical aspects influencing the accuracy and reliability of FEM simulations in AM is the incorporation of a volumetric heat source [30, 31]. The dummy volumetric heat source, first introduced by Goldak-Akhlaghi as a double ellipsoidal heat distribution [32], is used in the literature for more than two decades. A volumetric heat source reflects the spatial and temporal variations of heat input, accounting for the intricate interactions between the welding arc, the workpieces, and the surrounding environment [33]. However, it fails to accurately reflect the real thermophysical phenomena of welding. As is well-known in fusion welding heat is transferred as a flux across the weld pool's surface, with no volumetric heating occurring during the process. Fundamentally, the shape of the molten pool is the result of all governing equations acting together and ought not to be used as a pre-assumption of the model. By adding the fluid flow and temperature-dependent surface tension questions to the model, the real shape of the molten pool could be created, even by a flux-type heat source [34]. Taking into account these limitations, W. Ou et al. [35, 36] proposed an innovative solution to address these challenges by introducing an advanced finite difference control volume method. In this approach, the surface of the deposit is presumed to be flat during the computation of the thermal and velocity fields. In the following, the free surface profile of the deposit is approximated by minimizing the total surface energy on the upper surface of the deposit utilizing an adaptive mesh scheme. Eventually, the grids are modified to conform to the surface configuration, including the relocated thermal and velocity fields. Because gravity is factored into the energy formulation, the surface energy model can automatically adapt to various deposition scenarios and accurately depict all possible weld bead shapes [37]. However, the deposition behavior in overhanging surfaces such as thin-walls must be further studied. Despite all the limitations, FEM methods remain at the top when it comes to predicting microstructure evolution. Coupling microscopic phase-field modeling (PFM) and macroscopic FEM allows it to predict how the material's microstructure evolves under specific conditions in a multiscale model of WAAM [38].

Due to its computational efficiency and simplicity, particularly for straightforward geometries, the finite difference method (FDM) remains widely employed across

numerous fields of thermal and mechanical analysis, including computational fluid dynamics. Furthermore, ongoing advancements in FDM aim to mitigate the high computational costs associated with three-dimensional (3D) cases. A few studies have employed the FDM to investigate heat transfer, fluid flow, and electromagnetic field in the WAAM process [39–41]. In Fig. 1, a simple comparison is made between the output of the classical FEM and present FDM models. The previous studies are mainly limited to relatively medium to high currents ranges [40] along with the short simulation time [42] due to the noticeable governing equations participate in such multiscale modeling. In addition, some analyses were conducted with inaccurate presuppositions. For example, investigating the effect of the droplets' frequencies without considering thermal equilibrium with the heat flux and initial temperature of the droplets creates a significant error. Realistic and long-term simulation of WAAM is always challenging using the FDM and VOF methods mainly due to the enormous scale difference between the droplet size and the deposited layers, which needs a compromise between the minimum requirement mesh size (droplet radius) and consuming time. This research aims to provide essential pre-production information along with analytically predicting the feasibility of wall formation and its thickness using the FDM in the presence of temperature-dependent surface tension phenomenon. Finally, a model with long-term welding time and the low-range of current widely used in AM is presented. The results were validated through thin-wall samples fabricated from ER70S6 and ER316L wires.

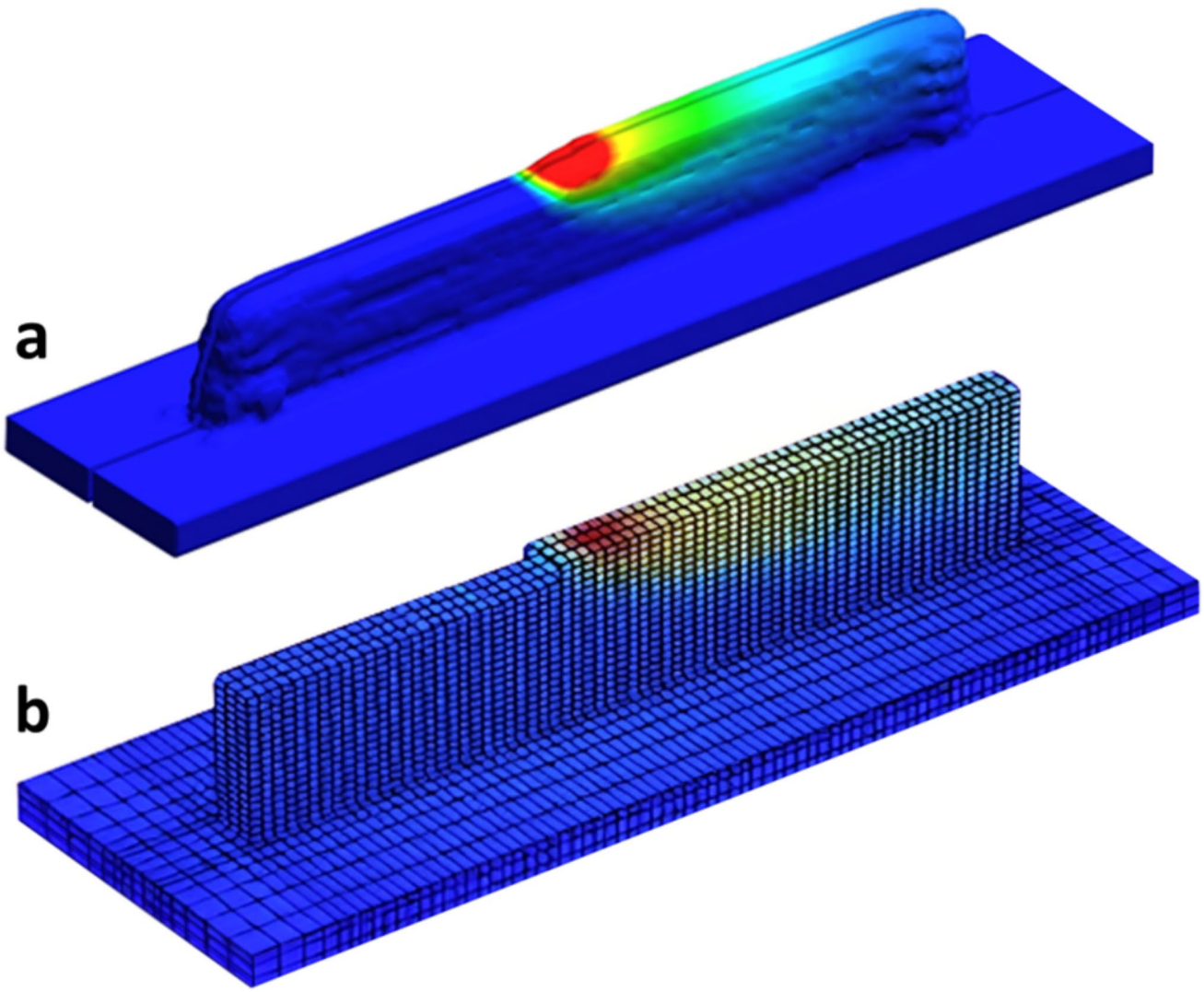


Fig. 1 Visualizing of FDM- based analyses results **a** and the classical FEM **b** adopted from [43]

2 Experimental procedure

Additive manufacturing was performed utilizing a Merkle PU520 DW welding power source through the P-GMAW process, which was integrated with a robotic carrier positioned horizontally, as illustrated in Fig. 2. The layers were deposited on an 8 mm thickness substrate using 1.2 mm diameter of solid wires. The chemical composition of filler wire and substrate are reported in Table 1. Welding variables included 125 A of nominal current (90 A effective) with DCEP polarity applied in a pulsed mode with a duty cycle of 30%, a frequency of 60 Hz, a voltage range of 16 to 22 V, and the average wire feed speed of 4.7 m/min. The temperature of the deposited layers was monitored using a thermal infrared (IR) camera, ensuring that the start point of each layer was air-cooled to 100 ℃ before adding the subsequent layer. To avoid excessive heat build-up and maintain consistent cooling conditions, a flexible dwell time ranging from 100 to 180 s was implemented. The contact tip-to-work distance (CTWD) was adopted 18 mm from the surface during build-up processes where the torch travel speed (TS) was set to 0.17 m/min based on a trial process. A shielding gas composed of 99% argon was applied at a flow rate of 15 L/min, primarily due to its ability to prevent spattering and promote rapid solidification. In each pass of layer deposition, approximately 1.8 mm is added to the height of thin-wall.

3 Mathematical formulation and modeling

3.1 Governing equations

A 3D model has been developed to analyze the shape of the weld pool and thin-walls formed by the WAAM process. The simulation domain was divided into fluid and void, with ambient air properties assigned to the void phase. Further, the FDM was employed to discretize and solve the heat transfer equation, which governs the temperature distribution and

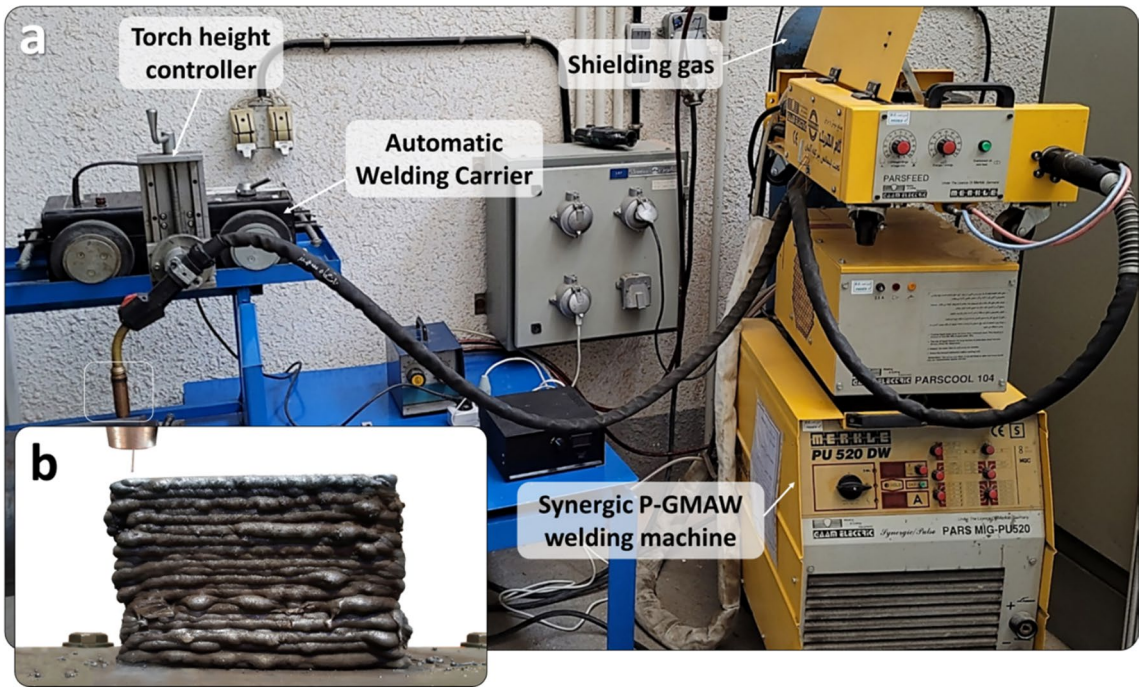


Fig. 2 **a** Merkle PU 520 DW equipment with a robotic carrier in a horizontal position, **b** the thin-wall of ER316L manufactured by 100 s of dwell time

Table 1 The composition of substrates and filler wires (wt.%)

Material	Cr	Ni	Mo	C	Si	Mn	P	S	Fe
St37	–	–	–	0.15	0.25	0.50	0.020	0.022	Bal
SS316	16.51	10.45	2.14	0.04	0.51	1.32	0.023	0.008	Bal
ER70S6	–	–	–	0.21	0.21	0.71	0.015	0.018	Bal
ER316L	17.32	12.04	2.36	0.02	0.33	1.89	0.045	0.005	Bal

the evolution of the solid–liquid interface during the melting process. At the same time, the VOF technique, according to Lagrangian principles, was utilized to track the evolution of the molten-pool free surface. Based on these assumptions, a total of four governing equations, including mass continuity, momentum (Navier–Stokes), energy conservation, and the VOF method, are solved sequentially as described in Eqs. (1) to (4) [26]. A list of variables is provided in Table 2.

$$\nabla.\vec{V} = \frac{\dot{m}_s}{\rho} \tag{1}$$

$$\frac{\partial \vec{V}}{\partial t} + \vec{V}\cdot\nabla\vec{V} = -\frac{1}{\rho}\nabla P + \frac{\mu}{\rho}\nabla^2\vec{V} - k_d\vec{V} + G_b + \frac{\dot{m}_s}{\rho}\cdot\left(\vec{V}_s - \vec{V}\right) \tag{2}$$

$$\frac{\partial h}{\partial t} + \vec{V}\cdot\nabla h = \frac{1}{\rho}\nabla.(K\nabla T) + \dot{h}_s \tag{3}$$

$$\dot{F}_s = \frac{\partial F}{\partial t} + \nabla\left(\vec{V}F\right) \tag{4}$$

where the term $k_d\vec{V}$ represents the flow losses in the porous media model. By neglecting the volume variation associated with the phase transformation [44], the porous media drag concept is used for modeling the flow in the transition zone according to Eq. (5) [45].

$$k_d = s_d \cdot \frac{f_s^2}{\left(1 - f_s\right)^3} \tag{5}$$

For non-isothermal phase transitions, the latent heat of fusion is accounted for in the enthalpy change between the solidus and liquidus temperatures, as shown in Eq. (6).

$$h = \begin{cases} C_s T & T \le T_s \\ h\left(T_s\right) + L_f\frac{T - T_s}{T_l - T_s} & T_s < T \le T_l \\ h\left(T_l\right) + C_l\left(T - T_l\right) & T_l \le T \end{cases} \tag{6}$$

3.2 Droplet model, body Forces, and boundary conditions

The beads' surface is subjected to arc heat, along with heat losses due to convection and radiation. The heat flux at the surface is mathematically represented by Eqs. (7) to (9):

Table 2 Symbols of mathematical formulation

Symbol	*Nomenclature*	*Symbol*	*Nomenclature*
C_l	Specific heat capacity* (liquidus, J/kg.K)	T_d	Droplet temperature (K)
C_s	Specific heat capacity* (solidus, J/kg.K)	T_l	Liquidus temperature (K)
d_t	Surface tangential direction (m)	T_m	Melting temperature of pure iron (K)
e_{S-B}	Stefan-Boltzmann constant (W/m^2.K^4)	T_0	Ambient temperature (K)
f_s	Local solid fraction	T_s	Solidus temperature (K)
f_d	Frequency of droplet generation (Hz)	$\vec{V}$	Velocity vector (m/s)
F	Volume fraction of fluid	$\vec{V}_s$	Mass source velocity vector (m/s)
$\dot{F}_s$	Change of F with the mass source	$\vec{V}_t$	Tangential velocity vector (m/s)
G_b	Total body force acceleration (m/s^2)	σ_q	effective radii of heat distribution
h	Enthalpy (J/kg)		
$\dot{h}_s$	Enthalpy source of droplet (W/kg)		
h_c	Convection coefficient (W/m^2.K)		
k_1	Entropy factor	*Greek symbol*	
k_d	Drag coefficient in porous model (1/s)	Γ_s	Surface excess at saturation (Mole/m^2)
K	Thermal conductivity* (W/m.K)	γ	Surface tension (N/m)
L_f	Latent heat of fusion (J/kg)	γ_m°	Surface tension of pure iron (N/m)
$\dot{m}_s$	Droplet mass source (kg/m^3.s)	ΔH°	Heat of adsorption (J/Mole)
$\vec{n}$	Normal vector of free surface	ε	Emissivity of the workpiece surface
Q	Heat power (J/s)	η	Arc thermal efficiency
r_d	Initial radius of the droplet (m)	η_d	Droplet thermal power per total arc power [46]
R	Gas constant (J/mol.K)	μ	Dynamic viscosity of fluid* (kg/m.s)
s_d	Drag coefficient constant (1/s)	ρ	Density of fluid* (kg/m^3)
t	Time (s)	$\rho_s.\rho_l$	Density of solid and liquid* (kg/m^3)
*	Temperature-dependent		

$$K\frac{\partial T}{\partial \vec{n}} = q - q_{conv.} - q_{rad.} \tag{7}$$

$$q_{conv.} = h_c\left(T - T_0\right) \tag{8}$$

$$q_{rad.} = \varepsilon e_{S-B}\left(T^4 - T_0^4\right) \tag{9}$$

Considering the efficiency of the P-GMAW process, η, the heat rate is assumed to be equal to the sum of the arc heat rate and the droplets ' heat content as Eq. (10) where the $\overline{UI}$ is the instantaneous average power [47].

$$\eta Q = \eta\left(Q_{arc} + Q_{drop}\right) \rightarrow \eta\overline{UI} = \eta\left(1 - \eta_d\right)\overline{UI} + \eta\eta_d\overline{UI} \tag{10}$$

The heat flux on the free surface is represented through a fixed Gaussian distribution model [48, 49] as:

$$q_{arc} = \frac{\eta\left(1 - \eta_d\right)UI}{\xi\pi\sigma_q^2} exp\left\{-\frac{\left(x - x_c - V_w t\right)^2}{\xi\sigma_q^2} - \frac{\left(y - y_c\right)^2}{\xi\sigma_q^2}\right\} \tag{11}$$

where the x_c and y_c are the moving coordinates of the arc center, and the V_w represents the welding velocity.

The specific heat capacity equation is utilized to determine the ideal amount of heat energy transfer from the droplets to the workpiece according to the frequency of droplet generation, f_d, represented by Eq. (12).

$$Q_{drop} = \frac{4}{3}\pi r_d^3 f_d\left\{\rho_l C_l\left(T_d - T_l\right) + \rho L_f + \rho_s C_s\left(T_s - T_0\right)\right\} \tag{12}$$

Utilizing previous studies, the molten pool shape is mainly governed by surface tension gradients, which are represented as Marangoni shear [34]. For a precise analysis of Marangoni shear, surface tension is formulated based on temperature and surface-active elements. In the case of ER70S6, due to the high amount of silicon and manganese in the filler metal composition and their deoxidant roles [50], the effect of oxygen is neglected, and only the effect of sulfur on the gradient of surface tension is considered. The surface tension relation of a pseudo-binary Fe-S system for ER70S6 were calculated using Eq. (13) [51]:

$$\gamma = \gamma_m^\circ - A\left(T - T_m\right) - RT\Gamma_s ln\left[1 + k_1 a_s e^{-\left(\Delta H^\circ / RT\right)}\right] \tag{13}$$

Note that the symbol A is equivalent to the negative gradient of surface tension for the pure iron and the activity of sulfur, a_s, was considered equal to the wt.% of the content.

Chromium content in SS316 plays a significant role in its physical properties, including surface tension. While specific studies on the direct relationship between chromium content and surface tension coefficient in SS316 are limited, Pichler et al. [52] reported surface tension measurement results obtained through electromagnetic levitation of SS316L. In this model, it was assumed that ER316L also follows similar physical properties to SS316L. The change in surface tension and the induced shear stress are balanced at the boundary condition as follows:

$$\mu\frac{\partial \vec{V}_t}{\partial \vec{n}} = -\frac{\partial \gamma}{\partial T}\frac{\partial T}{\partial d_t} \tag{14}$$

To determine the net surface tension force acting on a surface cell, the cell is considered as a control volume, and the net surface tension stresses acting at the sides of the cell are calculated. It was assumed that surface tension forces can be replaced by an equivalent surface pressure. Once all the force components have been evaluated, the net force in the direction normal to the surface is computed and divided by the area of the surface to obtain the equivalent surface tension pressure [45].

The buoyancy body force is adopted as the term G_b in the Navier–Stokes equation (Eq. (2) and is defined in the Cartesian coordinates in the following term:

$$\vec{F} = -\rho\vec{g}\beta_t\left(T - T_l\right) \tag{15}$$

The Boussinesq approximation utilized the coefficient of thermal expansion (β_t) to indirectly represent density variations resulting from temperature differences [17].

3.3 Numerical process and consideration

An incompressible fluid with phase-change capability is considered in the computational domain, incorporating the solid–liquid phase change for enhanced enthalpy calculation. The following additional assumptions are made to simplify the numerical model:

1. The molten metal is considered to be a Newtonian fluid that flow in a laminar mode.
2. Metal evaporation-induced volume loss in the liquid phase is assumed negligible.
3. Droplets are idealized as spherical and are considered to impinge upon the weld pool in uniformly.
4. The effect of plasma heat and electromagnetic pinch effect on droplets are incorporated by assuming an initial droplet temperature and velocity, respectively.
5. Based on prior research demonstrating that electromagnetic forces consistently exert less influence than Marangoni forces in determining volumetric forces [53], it is assumed that electromagnetic effects within the molten pool remain negligible across the applied low-current range.

To analyze the heat transfer and weld pool shape prediction, the commercial software FLOW-3D is utilized due to its high accuracy in tracking free-moving surfaces using the Lagrangian VOF method. The heat flux employs an in-house Fortran code compiled by the Intel® Fortran Compiler (ifort). Figure 3 presents a flowchart depicting the entire numerical procedure in this study.

Taking advantage of symmetry reduces the computational domain to just one-half of the total solution space. The domain was discretized using rectangular, 0.6 mm cuboid cells to capture the fusion zone accurately. Due to enhanced computational efficiency, the surrounding regions gradually increased in size to a maximum of 1.8 mm. The simulation domain is depicted schematically in Fig. 4. The droplets are generated at a distance of 1.5 mm above the surface and move vertically toward it with an initial velocity of 0.7 m/s at a temperature of 1900 K. Based on experimental data, the average increase in the simulated coordinate height per each pass was assumed to be 1.8 mm. Total efficiency of energy in P-GMAW is estimated to be around 68%, where 25% ($\eta\eta_d$) of the heat is transferred by the droplets, and the remaining 46% ($\eta(1-\eta_d)$) is directly imposed on the surface as a flux. Finally, the instantaneous average power ($\overline{UI}$) is calculated around 2165 J/s. The effective radius of the Gaussian heat was calculated based on the empirical equation as a function of the instantaneous current in SI units [17]:

$$\sigma_q = 6.205 \times 10^{-4} \times I^{0.28} \tag{16}$$

Governing equations for both heat transfer and fluid mechanics were implemented, focusing on the thermophysical properties of ER316L and ER70S6. The temperature-dependent physical properties of wires were described in Fig. 5 where the surface tension gradient exhibits significant changes with increasing temperature. All the other variables, including the welding parameters in the simulations were considered similar to those in the experimental samples. The solidus and liquidus temperatures of the ER70S6 were considered to be 1698 and 1778 K [17], respectively, while for ER316L, they were taken as 1675 and 1708 K [54], respectively.

The designed numerical model was examined for different conditions where the welding time of each pass takes 40 s, and a maximum total simulation time of 3400 s for 25 passes is defined in the time table (please see Table 3).

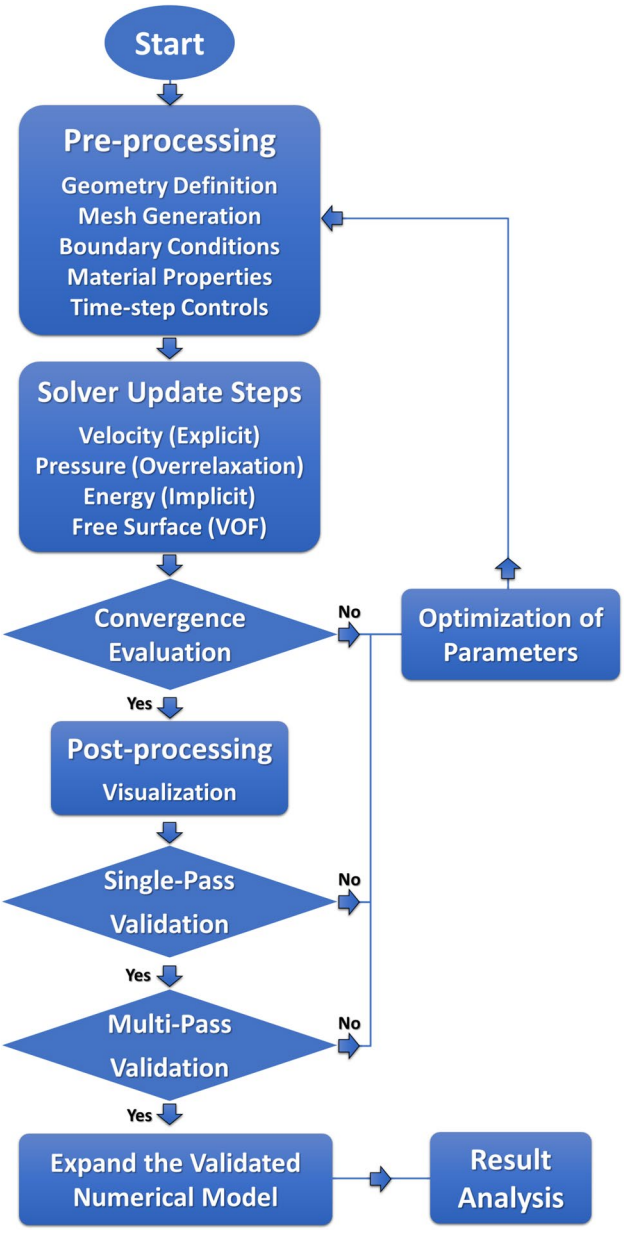


Fig. 3 A flowchart illustrating the simulation steps for the WAAM process

4 Results and discussion

4.1 Verification of the model

The model was evaluated using two experiments, S-B100 and SS-B100, produced via the bidirectional (B) strategy up to 25 passes (Fig. 6a and c). A central cross-section was selected as the verification candidate zone, as shown in Fig. 6b and d along with the longitudinal images. According to the findings, the experimental samples produced using the ER316 filler slightly widen in cross-section than those produced with the ER70S6 filler due to its lower thermal conductivity coefficient, especially in the initial passes where the wall height is negligible and the probability of heat accumulation is high. For a precise validation, the term of effective thickness is defined as a fully

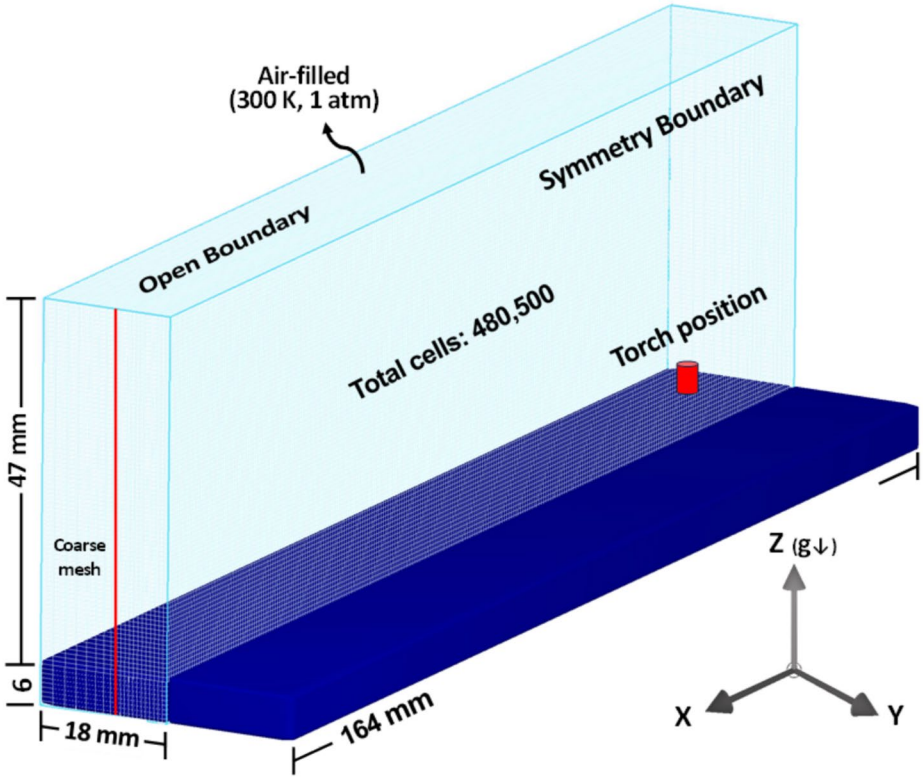


Fig. 4 A schematic of the simulation domain including the dimensions, mesh distribution, and the symmetric boundary

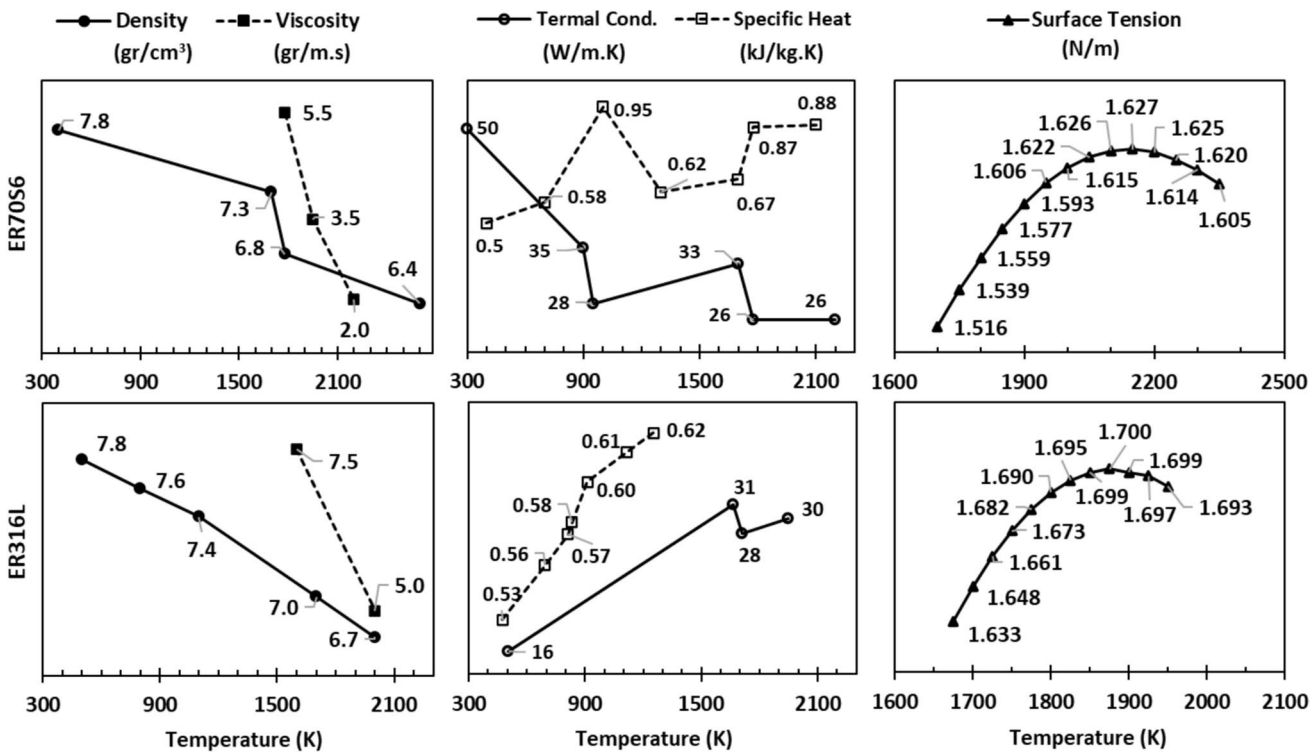


Fig. 5 The thermo-physical properties of ER70S6 [17, 51] and ER316L [52, 54]

Table 3 The timing and the production strategy of the experiments and the simulations

Sample NO	Filler Material	Manufacturing strategy	Dwell time	Target pass
S-U5	ER70S6	U*	5	10
S-B5		B*	5	10
S-U100		U*	100	10
S-B100		B*	100	10, 25
SS-U5	ER316L	U*	5	10
SS-B5		B*	5	10
SS-U100		U*	100	10
SS-B100		B*	100	10, 25

*U: Unidirectional, B: Bidirectional

preserved wall surface subsequent to machining, with all surface irregularities eliminated. Based on the simulation results for the S-B100 specimen, the effective thickness in the simulated model was measured to be 6.2 mm, which is 0.9 mm greater than that of the experimental specimen, indicating a 17% overestimation. It should be noted that excluding the upper third of the experimental specimen, which is characterized by considerable geometric deviations due to manufacturing inaccuracies, leads to a reduction in computational error by approximately 3%. In the case of the SS-B100 specimen, the model exhibited an effective thickness of 6.3 mm compared to 6.2 mm in the experimental measurement, reflecting a smaller error of 2% in the simulation results. Given the evidence, the model successfully replicated the experiments with an error margin below 5% in the total height measurements, demonstrating an acceptable level of conformity.

In the following analysis, the influence of the thermal conductivity coefficient and its disparity between the two alloys have been estimated in terms of their effect on temperature distribution. The temperature contour in the longitudinal profile well depicts the thermal effect of the 25th pass on the initial passes is greater in the ER70S6 sample than in the similar ER316 sample. This difference was displayed on the simulations with a threshold of 500 K (right side of Fig. 6b and d). As is clear from the simulation results (red zone in Fig. 6), the volume of the molten pool in the S-B100 sample is approximately 35 to 40% smaller than SS-B100, and this sample solidified faster, resulting in a greater tendency for the wall to increase in height relative to width. However, the difference in the height of the two samples, in the simulation and unlike the experiments, was found to be insignificant, and the role of the discussion above was reflected in a more regular form as shown in the cross-section of Fig. 6.

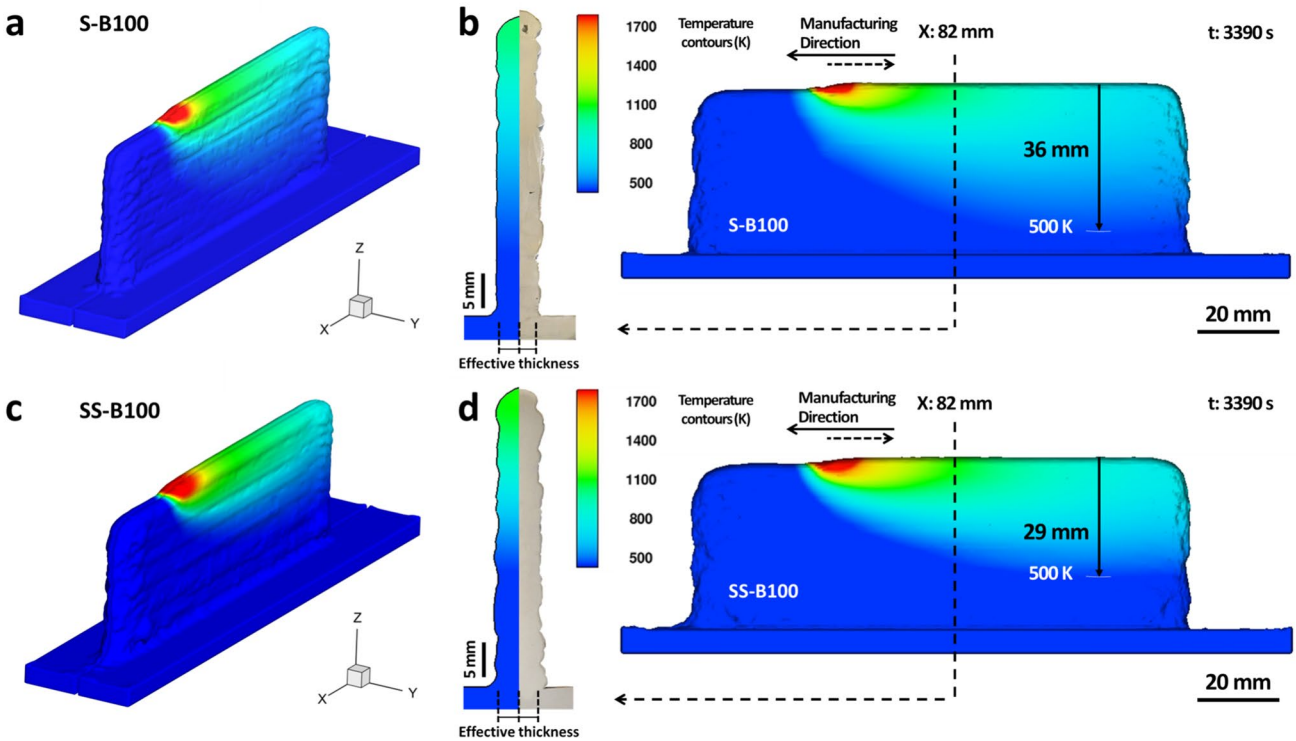


Fig. 6 WAAM process simulation results of the S-B100 (**a**, **b**) and SS-B100 (**c**, **d**) samples at 3390 s. **b** and **d** show the cross-section comparison with experiments and the longitudinal view in 2D modes

4.2 Macroscopic computed results

In this section, the macroscopic computed results were reported for each alloy composition and different manufacturing strategies. The first set includes models that were simulated using unidirectional (U) deposition by an interlayer dwell time of 5 s and 100 s, as shown in Fig. 7a, b and c, d, respectively. Note that the selection of candidates is based on the first macro sagging that occurred during the deposition of layers. In the case of insignificant collapse, the simulation was allowed to reach the target pass. It is found that the wall sagging takes place in both models at around 150 s (4th pass), while by extending the dwell time from 5 to 100 s, the models reach the target pass without sagging. Further, the S-U100 model shows a more uniform deposition geometry than SSU100 (Fig. 7c, d). This difference is also visible in the first layers of the experiments in Fig. 6, where the device alignment error is less.

The second set describes the models that were simulated in a bidirectional (B) condition using dwell times of 5 s and 100 s, as shown in Fig. 8a, b and c, d, respectively. A notable point is that sagging in the S-B5 model occurs during the fourth layer deposition with a dwell time of 5 s, while in SS-B5, sagging takes place earlier and after 119 s (3rd pass). This indicates a greater tendency for sagging in walls fabricated using ER316L wire in bidirectional deposition. Also, the higher heat conduction coefficient in the ER70S6 wire caused the S-U5 and S-B5 models to collapse in almost the same pass (140 to 150 s), and the effect of elevated heat accumulation in the bidirectional mode was delayed. By extending the dwell time from 5 to 100 s, the S-B100 and SS-B100 models reach the target pass without sagging similar to the unidirectional mode (Fig. 8c, d).

Temperature contours on the free surface show that the dimensions of the molten pool are expanded more in all ER316 than in ER70 results, regardless of the manufacturing strategy. Here, experimental validation was used to assess the simulation result of SS-B5. According to Fig. 8b in the experiment, the sagging occurs at the same pass and with a delay from the simulation indicated time zone. Although the sagging geometry is different from the simulation prediction, it is well known that the molten material at the end of the pass tends to flow into the space in front of the weld puddle more than the wall sides, and the delay between sagging times explains the different geometry.

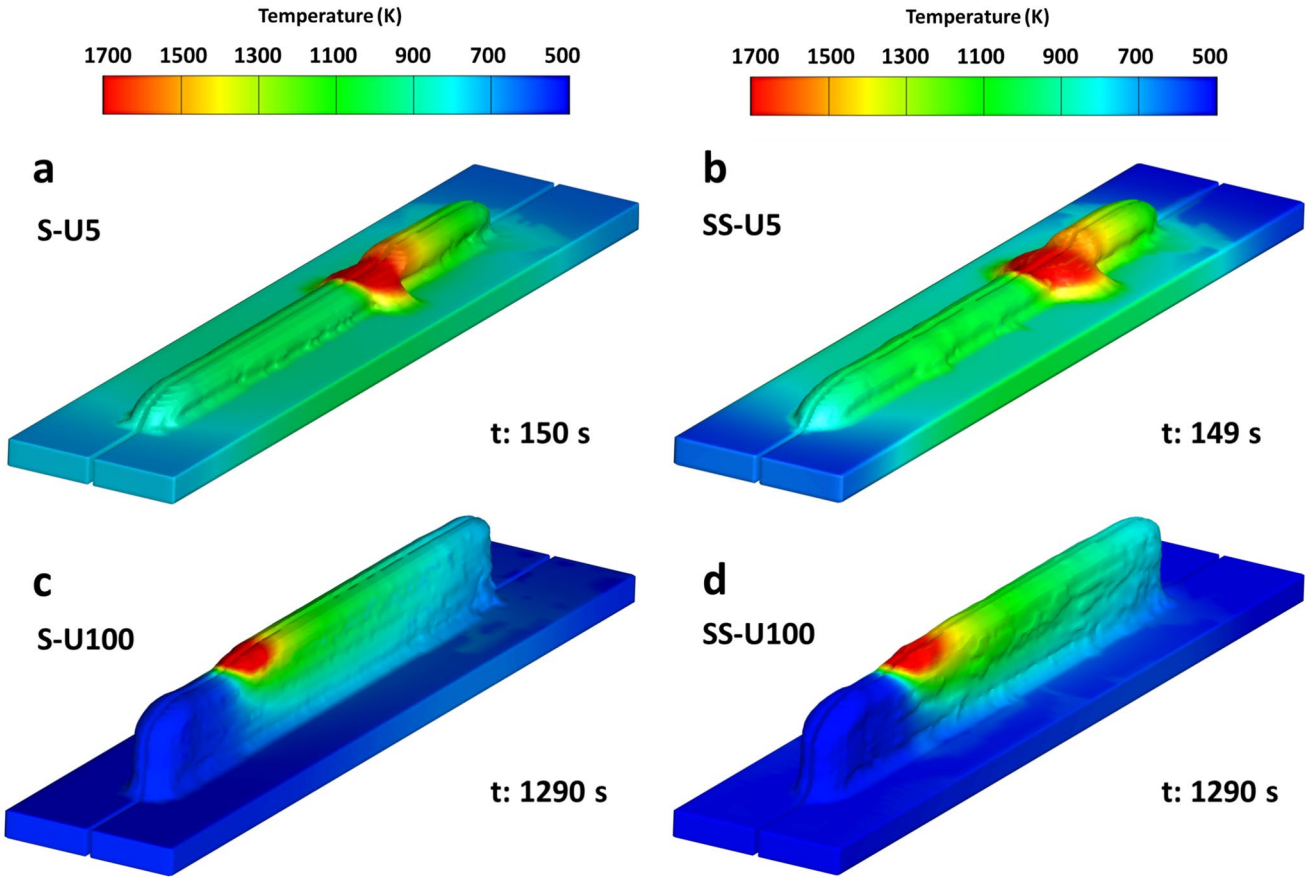


Fig. 7 The simulation results of the unidirectional deposition by dwell time of 5 s (**a**, **b**) and 100 s (**c**, **d**)

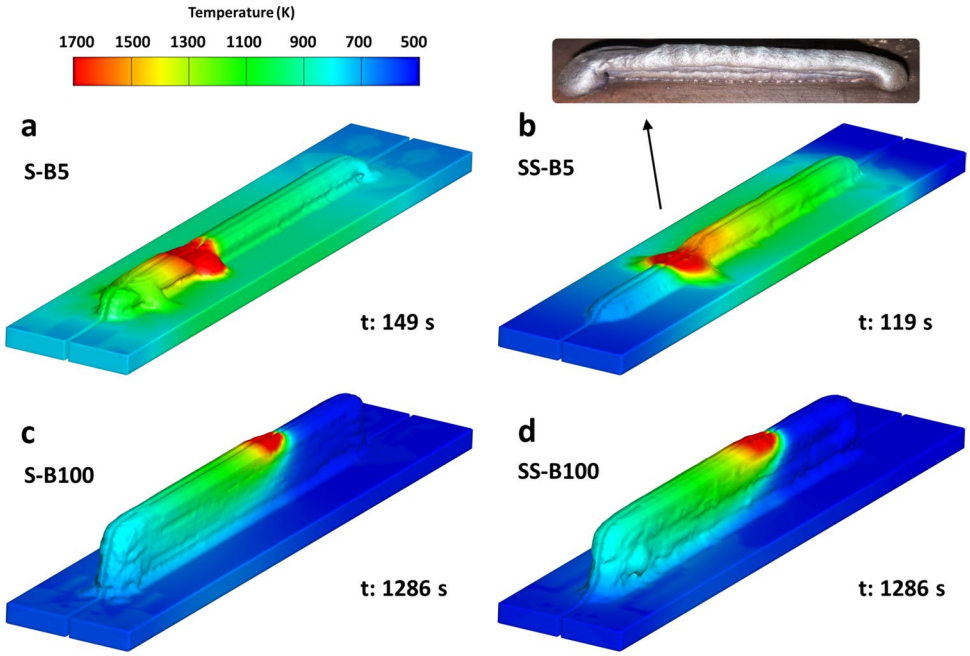


Fig. 8 The simulation results of the bidirectional deposition by dwell time of 5 s (**a**, **b**) and 100 s (**c**, **d**). In the case of SS-B5, experimental validation was used to assess the simulation results

4.3 Temperature analyses and the mechanism of sagging

Figure 9 demonstrates the average temperature change of the simulated thin-wall in a fixed dwell time and different deposition strategies. In all graphs, the effect of dwell time on the average temperature is observable as decreasing cycles before initiating the next pass. As shown in Fig. 9a, at the beginning of the third pass (with a dwell of 5 s), the difference between the average temperatures of the samples becomes severe, and subsequently, sagging takes place in all models before the end of the fourth pass, regardless of the manufacturing strategy chosen.

Despite the relationship between the rate of average temperature growth and the wall sagging, the graph itself cannot inherently predict the sagging time (indicated by the red arrows on the graph). By increasing the dwell time to 100 s, the average temperature fluctuations initially increase and after reaching a maximum, enter a relatively stable downward trend. Also, according to the graph, the range of average temperature changes in the bidirectional manufacturing strategy (Fig. 9b) is less than unidirectional mode.

For the analyses of temperature variations along the manufacturing rote, the point probes were employed at the middle and corner of the wall. The SS-B5, the most sensitive case, was a candidate where the probes were located at the center of the puddle (Y: 0) in the second and third passes. Note that the probes were designated as the Pass number-X position. Figure 10 illustrates how temperature fluctuates across various probes in the SS-B5 model.

Two distinct patterns can be observed in these graphs. The first type features a peak followed by a decline, with a brief period of stability at high temperatures before the subsequent drop. The short-lived stability at high temperatures is associated with molten pool oscillations before complete solidification, resulting from the continuous impact of filler droplets. The second type, which contains two peaks, corresponds to the sequence of deposition passes occurring generally in the lateral probes. Considering the sagging of the wall at 119 s, the temperature changes after this period are related to a layer with an irregular geometric shape compared to the main wall, and the analysis of temperature changes will not be reliable. Therefore, the temperature peaks before that are of interest. The notable point is that during the second pass, a temperature increase occurs as it traverses the central probe. However, in the third pass, the temperature exhibits a decline relative to the preceding corner probe. Given the importance of this result and the role of the probe position in analyzing temperature changes, the simulation was repeated in a limited domain, including three passes with finer meshes, and similar results were obtained. Accordingly, there seems to be a competition between heat accumulation due to a lower relative heat transfer coefficient and an increase in the area of heat dissipation, similar to what occurs in heat sinks. Though, the thin-wall continues to collapse, and this important question arises if heat dissipation overcomes heat accumulation in the 3rd pass, why did the collapse occur after the central probe. Prior to discussing this contradiction, the most important result of the present simulation is the introduction of a specific pass in which there is a possibility of collapse and instability, for

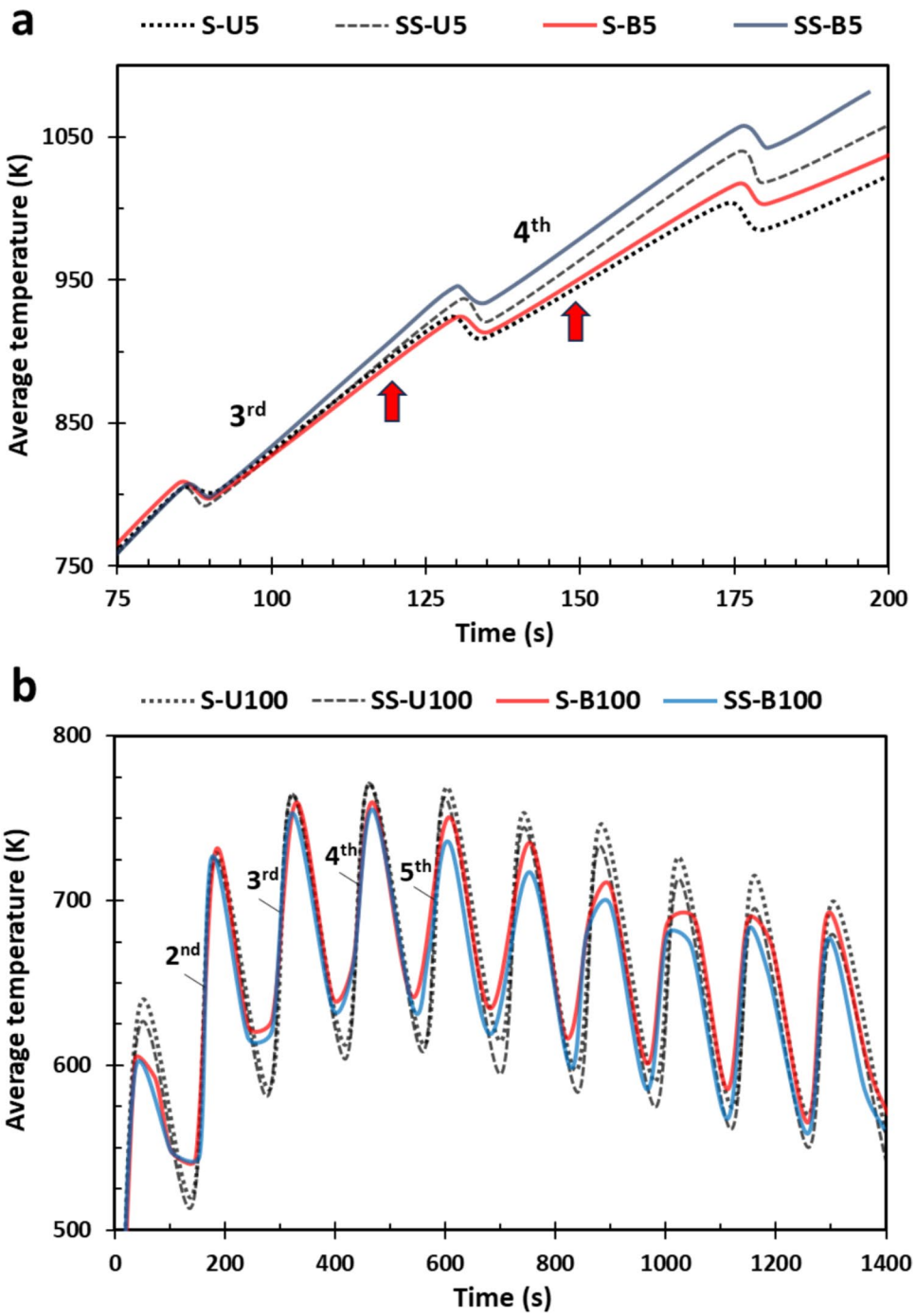


Fig. 9 Changes in average temperature of the simulated thin-wall in **a** 5 s and **b** 100 s of dwell time

example, the third pass of the SS-B5 and the fourth pass of the S-B5 model.

Two main reasons can explain the proposed contradiction. First, given the lack of complete uniformity in the layered area, minor dimensional changes in the previous layer thickness (as in the experimental sample) could have upset this partial balance and caused the wall to collapse. The second reason could be the role of the flow of materials inside the weld puddle without a significant effect on its temperature, such that the thin solidified layer around the molten pool melts due to these flows. In the following, the probability of wall sagging will be analyzed with this assumption. Figure 11 visualizes the temperature contours in the cross-section of the SS-B5 model precisely at the instant the wall starts to droop, along with the variations in surface tension pressure.

Despite extensive efforts to model and visualize surface tension at the instant of sagging, major challenges remain in this area. These challenges include the need for minimal time steps in numerical simulations and uncertainties

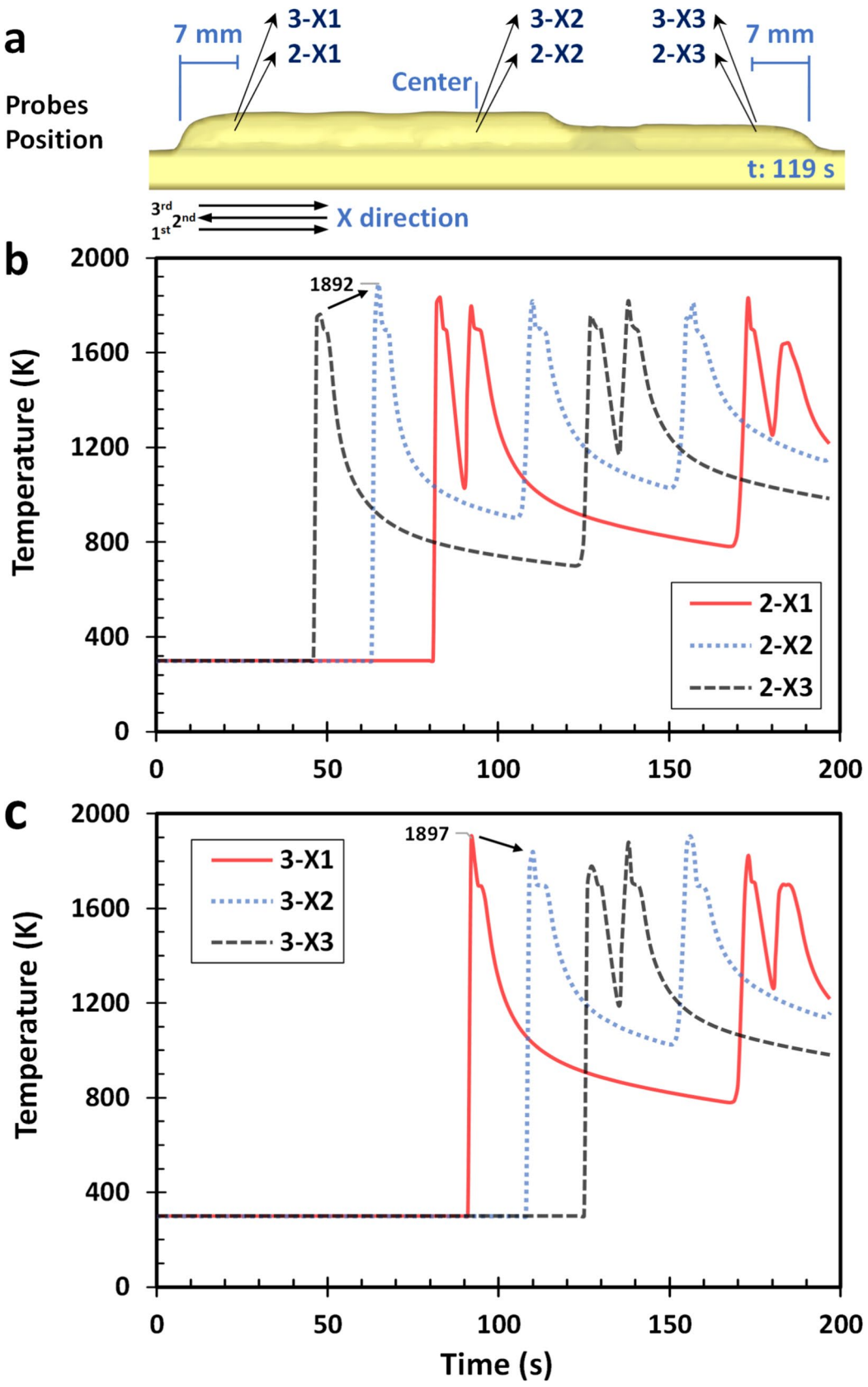


Fig. 10 A schematic of probes' position and codes in the SS-B5 model (**a**), the temperature change in the 2nd pass (**b**), and the 3rd pass through the probes (**c**)

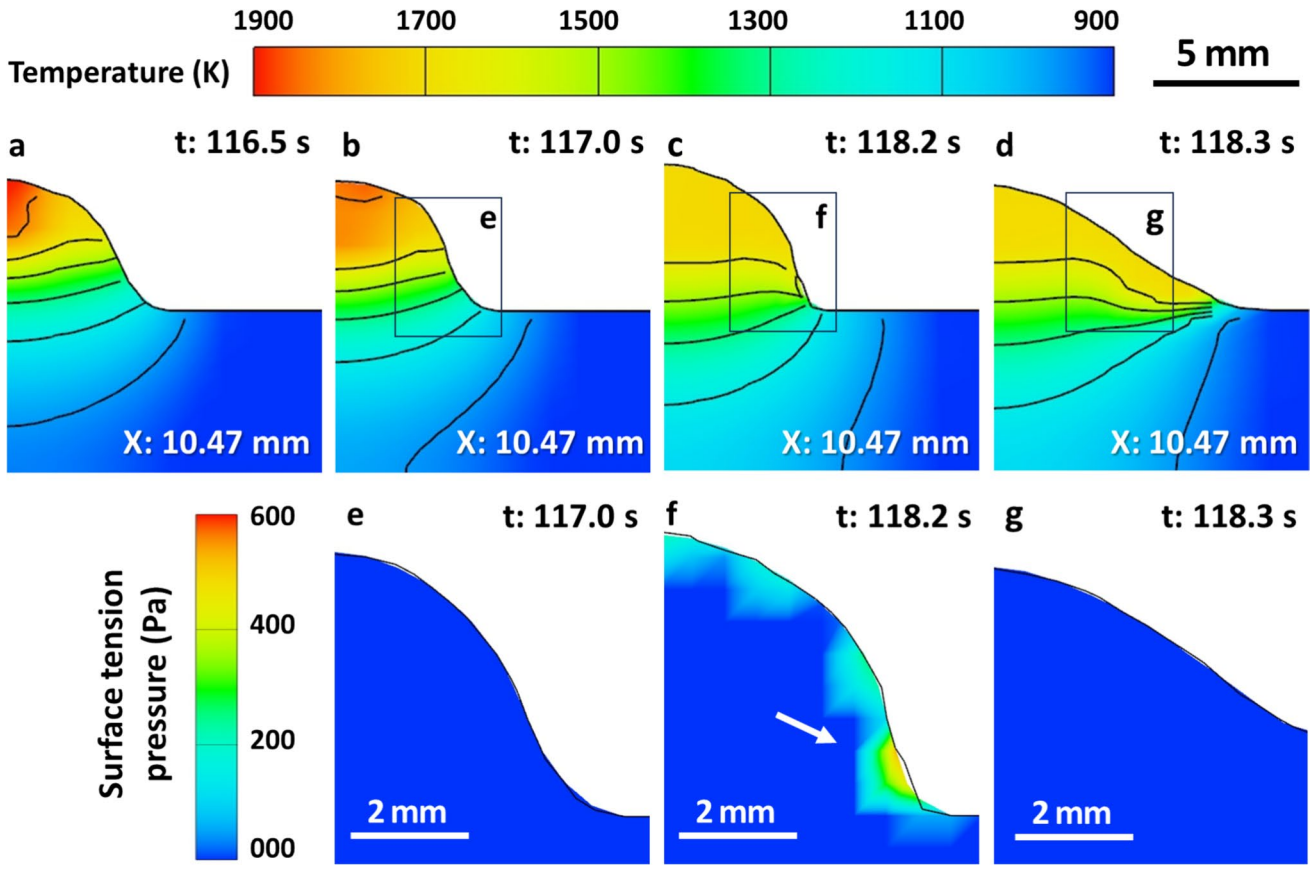


Fig. 11 Temperature contours in the cross-section of the SS-B5 model selected from the 3.rd pass (**a** to **d**) and the variation in surface tension pressure (**e** to **g**)

in determining the exact location where sagging initiates. These limitations have made it difficult to quantitatively calculate the critical state of sagging with sufficient accuracy. However, significant and sudden changes in surface tension pressure can be qualitatively observed and analyzed, particularly at the weld toes. These sudden changes can be considered indicators of the system approaching a critical state. According to Fig. 11a to d, as the torch passes, the central temperature decreases following the melting of the weld toe with a short delay, mainly due to fluid flow through the molten pool. The gradual shift of the 900 and 1100 temperature contours over time serves as a clear indicator of the model's low thermal conductivity. The change in puddle shape from finger type to cylindrical mode despite the temperature uniformity creates a significant increase in surface tension pressure near the weld toes (see Fig. 11c, f), and the sagging takes place as shown in Fig. 11d.

Simulation results show that the critical value of surface tension pressure, due to its simultaneous dependence on temperature and molten material viscosity, can be a candidate criterion for determining the instant of sagging. Given these discussions, analyzing the thin-wall sagging based solely on a single thermophysical parameter (such as thermal conductivity) will be associated with significant errors. This is due to the complexity of interactions between various parameters and their simultaneous impact on wall stability. Among these, temperature and surface tension pressure, as the comprehensive indicators, reflect the combined influence of various thermophysical parameters and can be used as a criterion for evaluating wall stability. Furthermore, developing high-precision experimental and numerical methods for accurately identifying the critical moment of sagging and the influencing parameters is of great importance.

5 Conclusion

The computational fluid dynamics model reliably predict the feasibility of thin-wall manufacturing with acceptable margin of error, even if the complex volumetric equations such as the effect of electromagnetic force are not fully included in the calculations. Furthermore, the proposed model is capable of estimating the susceptibility to sagging at a given pass; however, pinpointing the precise moment of sagging remains unfeasible due to the intricate interplay of influencing factors. Reducing the dwell time from 100 to 5 s

led to a higher probability of thin-wall sagging using both ER70S6 and ER316L wires. In unidirectional mode, each of the walls nearly collapsed by the 4th pass, whereas in bidirectional mode, the heat accumulation effect in ER316L accelerated sagging, causing failure by the 3rd pass. According to the simulation results the critical surface tension pressure, owing to its concurrent dependence on temperature and the viscosity of molten material, may serve as a potential criterion for identifying the manufacturing instability and possibility of sagging.

Author contribution HHZ: Writing – original draft, Visualization, Conceptualization, and Formal analysis. SAKM: Data Curation. VZL: Investigation and Resources.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

References

1. Queguineur A, Rückert G, Cortial F, Hascoët JY (2018) Evaluation of wire arc additive manufacturing for large-sized components in naval applications. Weld World 62:259–266. https://doi.org/10.1007/s40194-017-0536-8
2. Hirata Y (2003) Pulsed arc welding. Weld Int 17:98–115. https://doi.org/10.1533/wint.2003.3075
3. Zhong Y, Zheng Z, Li J, Wang C (2021) Fabrication of 316L nuclear nozzles on the main pipeline with large curvature by CMT wire arc additive manufacturing and self-developed slicing algorithm. Mater Sci Eng A 820:141539. https://doi.org/10.1016/j.msea.2021.141539
4. Chen X, Li J, Cheng X, He B, Wang H, Huang Z (2017) Microstructure and mechanical properties of the austenitic stainless steel 316L fabricated by gas metal arc additive manufacturing. Mater Sci Eng A 703:567–577. https://doi.org/10.1016/j.msea.2017.05.024
5. Gisario A, Kazarian M, Martina F, Mehrpouya M (2019) Metal additive manufacturing in the commercial aviation industry: a review. J Manuf Syst 53:124–149. https://doi.org/10.1016/j.jmsy.2019.08.005
6. Terrenoir L, Lartigau J, Arjunan A, Laguna Salvado L, Merlo C (2023) Influence of wire feed speed and torch speed on the mechanical properties of wire arc additively manufactured stainless steel. J Manuf Sci Eng. https://doi.org/10.1115/1.4063108
7. Rathee S, Srivastava M (2023) Wire arc additive manufacturing. CRC Press, Boca Raton. https://doi.org/10.1201/9781003363415
8. Mvola B, Kah P, Layus P (2018) Review of current waveform control effects on weld geometry in gas metal arc welding process. Int J Adv Manuf Technol 96:4243–4265. https://doi.org/10.1007/s00170-018-1879-z
9. de Sousa Figueiredo GG, de Mello Picchi IB, Lima dos Santos M, López EAT, Oliveira JP, de Abreu Santos TF (2023) Parametric study and response optimization for the wire + arc additive manufacturing of 316LSi via pulsed GMAW. Int J Adv Manuf Technol 129:3073–92. https://doi.org/10.1007/s00170-023-12470-2
10. Dos Santos EBF, Pistor R, Gerlich AP (2017) Pulse profile and metal transfer in pulsed gas metal arc welding: droplet formation, detachment and velocity. Sci Technol Weld Join 22:627–641. https://doi.org/10.1080/13621718.2017.1288889
11. Ríos S, Colegrove PA, Martina F, Williams SW (2018) Analytical process model for wire + arc additive manufacturing. Addit Manuf 21:651–657. https://doi.org/10.1016/j.addma.2018.04.003
12. Stavropoulos P, Foteinopoulos P (2018) Modelling of additive manufacturing processes: a review and classification. Manuf Rev. https://doi.org/10.1051/mfreview/2017014
13. Cho MH, Lim YC, Farson DF (2006) Simulation of weld pool dynamics in the stationary pulsed gas metal arc welding process and final weld shape. Suppl To Weld J 85(Suppl):271–283
14. Cho MH, Farson DF (2007) Understanding bead hump formation in gas metal arc welding using a numerical simulation. Metall Mater Trans B Process Metall Mater Process Sci 38:305–319. https://doi.org/10.1007/s11663-007-9034-5
15. Cheon J, Na S-J (2016) Influence of simulation methods of temperature distribution on thermal and metallurgical characteristics in GMA welding. Mater Des 108:183–194. https://doi.org/10.1016/j.matdes.2016.06.090
16. Wu D, Ishida K, Tashiro S, Nomura K, Hua X, Ma N et al (2023) Dynamic keyhole behaviors and element mixing in paraxial hybrid plasma-MIG welding with a gap. Int J Heat Mass Transf 200:123551. https://doi.org/10.1016/j.ijheatmasstransfer.2022.123551
17. Hamed Zargari H, Ito K, Kumar M, Sharma A (2020) Visualizing the vibration effect on the tandem-pulsed gas metal arc welding in the presence of surface tension active elements. Int J Heat Mass Transf 161:120310. https://doi.org/10.1016/j.ijheatmasstransfer.2020.120310
18. Ou W, Knapp GL, Mukherjee T, Wei Y, DebRoy T (2021) An improved heat transfer and fluid flow model of wire-arc additive manufacturing. Int J Heat Mass Transf. https://doi.org/10.1016/j.ijheatmasstransfer.2020.120835
19. Zhao W, Wei Y, Long J, Chen J, Liu R, Ou W (2021) Modeling and simulation of heat transfer, fluid flow and geometry morphology in GMAW-based wire arc additive manufacturing. Weld World 65:1571–1590. https://doi.org/10.1007/s40194-021-01123-1
20. Katopodes ND (2019) Chapter 12 - Volume of Fluid Method. In: Katopodes NDBT-F-SF, editor, Butterworth-Heinemann. https://doi.org/10.1016/B978-0-12-815485-4.00018-8.
21. Haidar J (1998) Predictions of metal droplet formation in gas metal arc welding. part II. J Appl Phys 84:3530–3540. https://doi.org/10.1063/1.368528
22. Hu J, Tsai HL (2007) Heat and mass transfer in gas metal arc welding. Part II: The metal. Int J Heat Mass Transfer 50:808–820. https://doi.org/10.1016/j.ijheatmasstransfer.2006.08.026
23. Cheon J, Kiran DV, Na SJ (2016) CFD based visualization of the finger shaped evolution in the gas metal arc welding process. Int J Heat Mass Transf 97:1–14. https://doi.org/10.1016/j.ijheatmasstransfer.2016.01.067
24. Barkhudarov M (2004) Lagrangian VOF Advection method for FLOW-3D. Flow Sci Inc. FSI-03-TN6:1–11.
25. Béraud N, Ledoux Y, Mechekour EH, Vignat F, Pourroy F (2024) Fast thermal simulation of WAAM processing: toward manufacturing strategy evaluation. CIRP J Manuf Sci Technol 55:234–246. https://doi.org/10.1016/j.cirpj.2024.10.007
26. Mu H, He F, Yuan L, Hatamian H, Commins P, Pan Z (2024) Online distortion simulation using generative machine learning models: a step toward digital twin of metallic additive manufacturing. J Ind Inf Integr 38:100563. https://doi.org/10.1016/j.jii.2024.100563
27. Montevecchi F, Venturini G, Grossi N, Scippa A, Campatelli G (2018) Idle time selection for wire-arc additive manufacturing: a

finite element-based technique. Addit Manuf 21:479–486. https://doi.org/10.1016/j.addma.2018.01.007
28. la Peña JAD-D, Limon-Leyva PA, Cortes P, Choo K (2025) Modeling of direct energy deposition for wire laser additive manufacturing. J Mater Eng Perform. https://doi.org/10.1007/s11665-025-10665-0
29. Aryal P, Choquet I (2024) Melt pool electromagnetic force model extended to account for free surface deformation – application to gas metal arc. Int J Heat Mass Transf 221:124987. https://doi.org/10.1016/j.ijheatmasstransfer.2023.124987
30. Unni AK, Vasudevan M (2021) Determination of heat source model for simulating full penetration laser welding of 316 LN stainless steel by computational fluid dynamics. Mater Today Proc 45:4465–4471. https://doi.org/10.1016/j.matpr.2020.12.842
31. Gupta DK, Mulik RS (2025) Numerical simulation and experimental investigation of temperature distribution during the wire arc additive manufacturing (WAAM) process. Prog Addit Manuf 10:631–645. https://doi.org/10.1007/s40964-024-00647-4
32. Goldak JA, Akhlaghi M (2005) Computational welding mechanics. Springer, US, Boston, MA
33. Kim YH, Gunasegaram DR, Cleary PW, Murphy AB (2025) Numerical modelling of wire arc additive manufacturing: methods, status, trends, and opportunities. J Phys D Appl Phys 58:143001. https://doi.org/10.1088/1361-6463/adb3b0
34. Hamed Zargari H, Ito K, Sharma A (2023) Effect of workpiece vibration frequency on heat distribution and material flow in the molten pool in tandem-pulsed gas metal arc welding. Int J Adv Manuf Technol 129:2507–2522. https://doi.org/10.1007/s00170-023-12424-8
35. Ou W, Mukherjee T, Knapp GL, Wei Y, DebRoy T (2018) Fusion zone geometries, cooling rates and solidification parameters during wire arc additive manufacturing. Int J Heat Mass Transf 127:1084–1094. https://doi.org/10.1016/j.ijheatmasstransfer.2018.08.111
36. Ou W, Knapp GLL, Mukherjee T, Wei Y, DebRoy T (2021) An improved heat transfer and fluid flow model of wire-arc additive manufacturing. Int J Heat Mass Transf 167:120835. https://doi.org/10.1016/j.ijheatmasstransfer.2020.120835
37. Glasder M, Fabbri M, Aschwanden I, Bambach M, Wegener K (2023) Towards a general and numerically efficient deposition model for wire-arc directed energy deposition. Addit Manuf 78:103832. https://doi.org/10.1016/j.addma.2023.103832
38. Gao X, Yao X, Wang Y, Wang H, Song G, Zheng Y et al (2025) Multiscale modeling of grain structure control in wire arc additive manufacturing of Ti6Al4V. J Mater Res Technol 35:3980–3993. https://doi.org/10.1016/j.jmrt.2025.02.019
39. Ogino Y, Asai S, Hirata Y (2018) Numerical simulation of WAAM process by a GMAW weld pool model. Weld World 62:393–401. https://doi.org/10.1007/s40194-018-0556-z
40. Li Y, Yun Z, Zhang W, Long X, Zhu J (2024) 3D modelling of layer-by-layer heat and mass transfer in wire arc additive manufacturing. Addit Manuf Front 3:200159. https://doi.org/10.1016/j.amf.2024.200159
41. Akbarzadeh Chiniforoush E, Zargari HH, Jandaghi MR, Moverare J, Warsi R, Gür CH (2025) A sustainable strategy for wire arc additive manufacturing of high-performance duplex stainless steel: microstructural refinement and mechanical anisotropy reduction. Mater Sci Eng A 943:148785. https://doi.org/10.1016/j.msea.2025.148785
42. Huang J, Li Z, Yu S, Yu X, Fan D (2022) Real-time observation and numerical simulation of the molten pool flow and mass transfer behavior during wire arc additive manufacturing. Weld World 66:481–494. https://doi.org/10.1007/s40194-021-01214-z
43. Nain V, Carin M, Engel T, Boisselier D, Cordier C (2020) Numerical modeling of wire directed energy deposition additive manufacturing (Wire-DED) process. Comsol Conf North Am Online Event, Fr. octobre.
44. Leffler B, Malm S (1977) Volume changes accompanying solidification of some austenitic stainless steels. Met Technol 4:81–90. https://doi.org/10.1179/030716977803292457
45. FLOW-3D user's manual. Inc, Flow Sci 2018;v11.2.
46. Wang Y, Tsai HL (2001) Impingement of filler droplets and weld pool dynamics during gas metal arc welding process. Int J Heat Mass Transfer 44:2067–2080. https://doi.org/10.1016/S0017-9310(00)00252-0
47. Joseph A, Harwig D, Farson DF, Richardson R (2003) Measurement and calculation of arc power and heat transfer efficiency in pulsed gas metal arc welding. Sci Technol Weld Join 8:400–406. https://doi.org/10.1179/136217103225005642
48. Cho DW, Na SJ, Cho MH, Lee JS (2013) Simulations of weld pool dynamics in V-groove GTA and GMA welding. Weld World 57:223–233. https://doi.org/10.1007/s40194-012-0017-z
49. Hu J, Guo H, Tsai HL (2008) Weld pool dynamics and the formation of ripples in 3D gas metal arc welding. Int J Heat Mass Transf 51:2537–2552. https://doi.org/10.1016/j.ijheatmasstransfer.2007.07.042
50. Ahsan MRU, Cheepu M, Ashiri R, Kim TH, Jeong C, Do PY (2017) Mechanisms of weld pool flow and slag formation location in cold metal transfer (CMT) gas metal arc welding (GMAW). Weld World 61:1275–85. https://doi.org/10.1007/s40194-017-0489-y
51. Sahoo P, Debroy T, McNallan MJ (1988) Surface tension of binary metal-surface active solute systems under conditions relevant to welding metallurgy. Metall Trans B 19:483–491. https://doi.org/10.1007/BF02657748
52. Pichler P, Leitner T, Kaschnitz E, Rattenberger J, Pottlacher G (2022) Surface tension and thermal conductivity of NIST SRM 1155a (AISI 316L stainless steel). Int J Thermophys 43:1–17. https://doi.org/10.1007/s10765-022-02991-5
53. Cho W-I, Na S-J (2021) Impact of driving forces on molten pool in gas metal arc welding. Weld World 65:1735–1747. https://doi.org/10.1007/s40194-021-01138-8
54. Pichler P, Simonds BJ, Sowards JW, Pottlacher G (2020) Measurements of thermophysical properties of solid and liquid NIST SRM 316L stainless steel. J Mater Sci 55:4081–4093. https://doi.org/10.1007/s10853-019-04261-6

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